

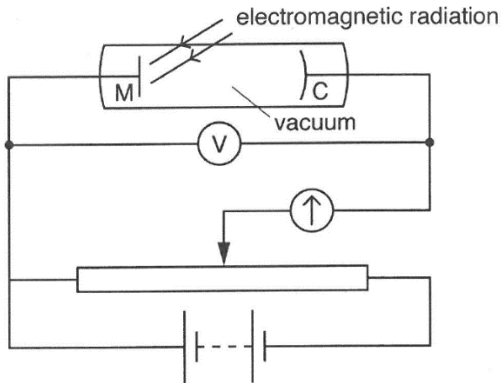
Q8(a)(i)

Photoelectric Effect: when light of a high enough frequency is shone on a metal surface, electrons are emitted.

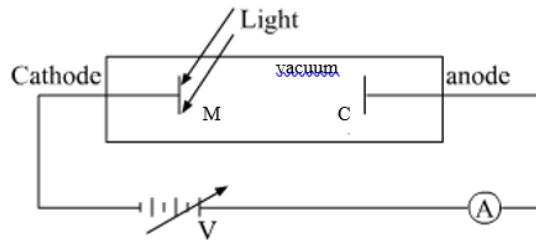
8(a)(ii) [Any 2]

- (1) There is a minimum frequency of radiation required, below which photoelectric emission does not occur.
- (2) There is no time lag between the illumination of the metal surface by energetic radiation and the emission of photoelectrons,
- (3) The maximum kinetic energy of the photoelectron is dependent on the frequency of light but independent of the intensity of the radiation.

8(a)(iii)



Or



Apply a variable p.d. V across 2 plates, the emitter (M) and collector (C). C is negative with respect to M. Start with $V = 0$, and shine the light through the window at M. As electrons are emitted by M and collected by C, the ammeter will register a current.

Now increase the retarding p.d. V until the ammeter just shows zero current. The p.d. that just causes the current to drop to zero is the stopping potential V_s . Then, maximum kinetic energy of electrons is $= eV_s$, where e = charge of 1 electron.

8(b)

There is kinetic energy of the electron moving around the nucleus of the atom and the electrical potential energy of interaction of the electron with the nucleus.

8(c)(i)1.

From point A, emission line spectrum from the hydrogen will be observed.

The cooler hydrogen will absorb parts of the spectrum of white light and reemit in all directions. Some of the coloured lines will be seen at A

8(c)(i)2.

At point B, an absorption line spectrum from the hydrogen will be observed.

This is because some specific spectral wavelengths from the hot gas will be absorbed by the cooler hydrogen. Hence the full colour spectrum of white light will have missing wavelengths represented by a set of dark lines.

8(c)(ii)1.

Transition from $n = 4$ and $n = 2$: $\Delta E = hf$

$$\Rightarrow (3.40 - 0.85) \times 1.6 \times 10^{-19} = 6.63 \times 10^{-34} \times f \quad \Rightarrow f = \mathbf{6.15 \times 10^{14} \text{ Hz.}}$$

8(c)(ii)2.

$$(E_4 - E_2) = (E_4 - E_3) + (E_3 - E_2).$$

$$\text{Thus, } hf_{42} = hf_{43} + hf_{32} \Rightarrow f_{42} = f_{43} + f_{32}.$$

$$\text{i.e. } 6.15 \times 10^{14} = 1.60 \times 10^{14} + f_{21}$$

$$\Rightarrow f_{21} = 4.55 \times 10^{14} \text{ Hz.}$$

$$\text{Hence, the emitted wavelength, } \lambda_{21} = c/f_{21} = \mathbf{6.59 \times 10^{-7} \text{ m.}}$$

8(d)(i)

$$p = h/\lambda$$

$$1.66 \times 10^{-27} \times 1.0 \times 10^5 = h/\lambda$$

$$\lambda = \mathbf{3.99 \times 10^{-12} \text{ m}}$$

8(d)(ii)

For diffraction to be observable, the wavelength must be comparable to the dimension of the aperture the wave is passing through.

The wave nature of electrons can be observed in electron diffraction experiments because the typical inter-atomic separation in crystals is about 10^{-10} m which is comparable to that of the de Broglie wavelength of the free electron travelling at the speed of $1.0 \times 10^5 \text{ m s}^{-1}$.

It is difficult to observe diffraction effects involving neutrons since their wavelengths are much shorter compared to the inter-atomic separation of crystals.